Chapter 2: Prompt Engineering & AI Content Creation

Class 12: Practical / Activities (10 Minutes)

Topic: Prompt Engineering Practice, Revision & Practical Assessment

---

Activity 1: Prompt Engineering Practice & Revision (4 Minutes)

Objective

To help students revise and apply all prompt engineering concepts learned in this chapter by creating, testing, refining, and evaluating AI prompts for different academic and creative tasks.

Materials Required

- Computer/Laptop or Tablet
- Internet Connection
- AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, or Perplexity AI)
- Student Practical Worksheet
- Notebook & Pen
- Teacher Observation Sheet

Instructions

1. Open any AI tool.
2. Select **one topic** from the list provided.
3. Complete the following tasks:
    - Write an effective research prompt.
    - Write an image generation prompt.
    - Write a presentation prompt.
    - Optimize one of the prompts by adding more details.

4. Generate AI responses for each prompt.
5. Compare the responses and identify the best result.
6. Organize all prompts and responses in the worksheet.

Suggested Topics

- Artificial Intelligence
- Renewable Energy
- Climate Change
- Cyber Security
- Water Conservation
- Space Exploration
- Healthy Lifestyle
- Smart Education
- Digital Citizenship
- Entrepreneurship

Sample AI Prompt

Research Prompt

"Act as a Science teacher. Explain Renewable Energy for Class X students with key features, advantages, disadvantages, and examples."

Image Prompt

"Create a modern educational poster on Renewable Energy with wind turbines and solar panels, flat illustration style, blue and green color theme, bright lighting, 4K quality."

Presentation Prompt

"Prepare a 6-slide presentation on Renewable Energy with slide titles, key points, speaker notes, visuals, and a conclusion suitable for Class X students."

Student Practical Worksheet

Task	Completed (✓)
Topic Selected	☐

Task	Completed (✓)
Research Prompt Written	☐
Image Prompt Written	☐
Presentation Prompt Written	☐
Prompt Optimized	☐
AI Responses Generated	☐
Final Review Completed	☐

Student Activity Checklist

Students should:

✔ Select one topic.

✔ Write three different AI prompts.

✔ Optimize one prompt.

✔ Generate AI responses.

✔ Compare the quality of responses.

Teacher Observation Checklist

☑ Writes structured prompts.

☑ Demonstrates prompt engineering skills.

☑ Applies optimization techniques.

☑ Reviews AI responses carefully.

Activity 2: Practical Assessment (3 Minutes)

Objective

To assess students' ability to independently apply prompt engineering skills for research, creative content generation, and professional communication.

Materials Required

- AI Tool
- Practical Assessment Sheet
- Teacher Evaluation Sheet

Instructions

Students should independently complete the following practical tasks:

1. Write an effective AI prompt.
2. Generate an AI response.
3. Improve the prompt.
4. Generate the refined response.
5. Explain the improvement.
6. Submit the completed worksheet.

Practical Assessment Worksheet

Assessment Task	Completed (✓)
Prompt Written	☐
AI Response Generated	☐
Prompt Improved	☐
Refined Response Generated	☐
Comparison Completed	☐
Final Submission	☐

Discussion Questions

1. Which prompt produced the best AI response?
2. Why is prompt optimization important?
3. How did AI help complete your task?
4. Which prompt engineering technique was most useful?

Teacher Observation Checklist

- [x] Demonstrates independent prompt writing.
- [x] Applies prompt optimization.
- [x] Evaluates AI responses effectively.
- [x] Uses AI responsibly.

Activity 3: Self-Assessment & Chapter Reflection (3 Minutes)

Objective

To help students reflect on their learning, evaluate their prompt engineering skills, and identify areas for future improvement.

Materials Required

- Reflection Worksheet
- Teacher Observation Sheet

Instructions

1. Complete the self-assessment checklist.
2. Reflect on your learning throughout the chapter.
3. Write one strength and one area for improvement.

4. Share one real-life application of prompt engineering.

Reflection Worksheet

Answer the following questions:

1. Which prompt engineering skill improved the most?

---

2. Which activity did you enjoy the most?

---

3. How can prompt engineering help in future studies or careers?

---

4. What responsible AI practice will you always follow?

---

5. Which prompt engineering technique would you like to improve further?

Self-Assessment Checklist

Skill	Yes	Needs Practice
I can write effective AI prompts.	☐	☐
I can optimize and refine prompts.	☐	☐
I can generate research and creative content.	☐	☐
I can evaluate AI responses.	☐	☐
I understand responsible AI use.	☐	☐
I can apply prompt engineering in real-life tasks.	☐	☐

Teacher Observation Checklist

- ☑ Completes self-assessment honestly.
- ☑ Demonstrates confidence in prompt engineering.
- ☑ Reflects thoughtfully on learning.
- ☑ Participates actively.

Mini Practical Assessment Rubric

Assessment Area	Marks
Prompt Writing & Structure	20
Prompt Optimization & Refinement	20
AI Response Evaluation	20
Practical Application & Problem Solving	20
Reflection, Participation & Responsible AI Use	20
Total	**100 Marks**

Teacher Observation Checklist

- ☑ Demonstrates proficiency in writing and refining AI prompts.
- ☑ Uses prompt engineering techniques to generate accurate and high-quality AI responses.
- ☑ Evaluates and improves AI-generated content through critical analysis.
- ☑ Applies responsible AI practices, ethical use, and academic integrity.
- ☑ Completes all practical activities confidently and actively participates in discussions.

Practical Time Distribution

Activity	Duration
Prompt Engineering Practice & Revision	**4 Minutes**
Practical Assessment	**3 Minutes**
Self-Assessment & Chapter Reflection	**3 Minutes**
Total Practical Time	**10 Minutes**

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Apply the complete **prompt engineering workflow**, including **prompt creation, optimization, refinement, evaluation, and improvement**.
- Design effective AI prompts for **research, image generation, presentations, reports, business communication, and creative content creation**.
- Evaluate AI-generated responses for **accuracy, relevance, organization, completeness, and quality**.
- Demonstrate independent problem-solving by refining prompts to achieve better AI outputs.
- Practice **responsible, ethical, and safe use of AI** while creating academic and professional content.
- Successfully demonstrate readiness for the **Chapter-End AI Prompt Challenge, Practical Assessment, and future AI-assisted learning and workplace applications**.